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ALGORITHM FOR LOCAL LOOP OPTIMIZATION OF MULTISTAGE FLOTATION PROCESSES

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Abstract: This paper discusses the application of the Local Loop Optimization algorithm to improve the parameters of a multistage flotation process. The basic idea is to adjust process parameters such as reagent feed rates, equipment volumes and others to maximize the yield of valuable minerals and minimize losses. The technique involves iterative application of the Local Loop Optimization algorithm to process models based on physical and chemical principles of flotation. Initial approximations for the parameters are taken from experimental data or previous experiments. The Local Contour Optimization algorithm is then used to find local optimums by varying the parameters and evaluating their effect on the output. The results of the study show that the application of the Local Contour Optimization algorithm can significantly improve the efficiency of the flotation process by increasing the yield of valuable minerals and reducing losses. This approach provides a reduction in production costs and increases the competitiveness of the enterprise in the mining industry.

Key words: flotation, local loop optimization (LCO), optimization algorithms, multi-stage processes (MSP), parameter optimization, loop control, optimization models, local optimization methods, automation of flotation processes

INTRODUCTION. The increasing needs of the national economy in non-ferrous metals and the deterioration of the raw material base cause the need not only to increase the volume of mining and processing of ore, but also to further improve the technique and technology of non-ferrous metal ores enrichment on the basis of modern trends in its development, achievements, experience of advanced domestic and foreign enterprises, the use of automation and computer technology to control technological processes. Local contour optimization of multistage flotation processes is an important area of research in the mining industry. Local contour optimization (LCO) algorithm is an effective tool for optimization of multistage flotation processes in mining industry. Flotation is the main process used to beneficiate ores of valuable minerals by adhering them to air bubbles in an aqueous medium. Optimization of this process is critical to improve efficiency and reduce costs. Intensification of flotation process operation, increasing its efficiency is an important technical and economic problem. One of the most important ways to solve this problem is the improvement of flotation control based on the application of modern methods of system analysis, mathematical modeling, computer technology and automation.

Industrial flotation complex is a set of slurry-air mixture flows, technological separation process and apparatuses for its implementation. Effective management of the complex should ensure the optimization of the conditions of the flotation process in accordance with the accepted criteria at all its stages. Achievement of this goal is possible when providing control of the flotation complex as a whole, including interconnected operational regulation of the parameters of the technological scheme, characterizing the material flows of pulp, and the parameters of the technological mode of flotation. At present, due to the lack of theory of flotation as an object of control, due to the complexity of processes, specificity of each of them at different enterprises, and for a number of other reasons, the work to improve the management of flotation processes are carried out piecemeal, do not cover the problems of flotation management as a whole and are reduced, as a rule, to solving individual issues of optimization of technological modes or flotation schemes.

One of the important tasks of the analysis of SME functioning is the identification and formation of criteria for optimization of technological processes and their schemes, quantitative assessment of their relationships with the parameters of raw materials and production products.

Any multistage system as a complex technical system has a functional, technological, organizational and information structure, the optimization of the functioning of which is achieved on the basis of a system of criteria of different nature, characterizing the degree of achievement of private goals for each alternative.

MATERIAL AND METHODS. In the formulation of any SME management problem, the control criterion is a functional linking the main technical and economic indicators of production with control actions under given constraints.

Let us consider the optimization problem formulated as follows. On the basis of the obtained mathematical description of the object activity it is required to find the maximum achievable in a given period of time output characteristics of the object and to determine the best conditions of the object functioning.

To solve the control problem of multistage flotation process, first of all, it is necessary to have an adequate mathematical model of the optimized process, the general form of which is expressed by the relation

$$\beta = f(\bar{x}, \bar{u}, \bar{\xi}, \bar{a}),$$

where β is the process output; are $\bar{x}, \bar{u}, \bar{\xi}$ vectors of controlled, control, and uncontrolled parameters; $\bar{a}$ is the coefficient of the model.

When solving the optimization problem, i.e. the problem of determining the extreme value of the output indicator, the optimality criterion is often considered as a function of control and controlled parameters. In this case, the solution to the optimization problem of a complex multistage flotation process (circuit) is to determine the optimal values of control parameters providing the extremum of the function.

Translated with DeepL.com (free version)When solving the optimization problem, i.e. the problem of determining the extreme value of the output indicator β, the optimality criterion is often considered as a function of control and controlled $\bar{x}$ parameters $\beta = \beta(\bar{x}, \bar{u}, \bar{a})$. In this case, the solution to the optimization problem of a complex multistage flotation process (circuit) is to determine the optimal values of the control parameters $\bar{\mathcal{U}} = (\mathcal{U}_1^*, \mathcal{U}_2^*, \dots, \mathcal{U}_n^*)$, providing the extremum of the function

$$\beta(\bar{x}, \bar{u}, \bar{a}) = \underset{\bar{u} \in \mathcal{D}}{max} \beta(\bar{x}, \bar{u}, \bar{a})$$

if the condition

$$\mathcal{D} = \left\{ \bar{u} \in \frac{\mathcal{R}^{\pi}}{c_i} \le u_i \le d_i, i = 1,2, \dots, n \right\}$$

and stable values $x_1, x_2, \dots, x_m$.

When solving optimization problems, it is either impossible or extremely difficult to determine exactly the values of the optimal parameter $\bar{\mathcal{U}}^*$. Therefore, the optimal value of $\bar{\mathcal{U}}^*$ is found by approximate methods [1,2].

The choice of one or another method is largely determined by the problem statement, as well as by the mathematical model of the optimization object used.

In general, methods for solving optimization problems can be divided into analytical and algorithmic methods: the former are those that solve the optimization problem by means of a formula (i.e., quite accurately and in a finite number of steps); algorithmic methods are based on the idea of successive approximation.

When solving optimal problems in ACS multistage processes, algorithmic methods are used in the majority of cases, as it specifies the method of transition from one approximation of the vector of optimized parameters$\mathcal{X}_{\mathcal{N}}$ to another $\mathcal{X}_{\mathcal{N}+1}$. The necessary condition is the convergence of the method to the exact solution of the problem $\mathcal{X}^*$

$$\lim_{\mathcal{N}\to\infty} \mathcal{X}_{\mathcal{N}} = \mathcal{X}^*. \quad (1)$$

In the simplest case, an algorithmic method is given by the operator $\mathcal{F}$, which connects two successive approximations by a recurrence formula

$$\mathcal{X}_{\mathcal{N}+1} = \mathcal{F}(\mathcal{X}_{\mathcal{N}}). \quad (2)$$

In more complex cases, the application of algorithmic methods requires knowledge of a large amount of background:

$$\mathcal{X}_{\mathcal{N}+1} = \mathcal{F}(\mathcal{X}_{\mathcal{N}}, \mathcal{X}_{\mathcal{N}+1}, \dots, \mathcal{X}_{\mathcal{N}+n}). \quad (3)$$

Since the problem of optimization of the flotation process operation comes down to maximizing the yield of the finished product presented in a nonlinear form at a given quality of the finished concentrate and taking into account the restrictions on circulation flows, it is advisable to solve it by methods of nonlinear programming [1,3,5].

The optimization algorithm of the random search method. In the process of searching for a local minimum, the method of determining the best direction of descent with self-learning is used as an efficiency criterion, the essence of which is as follows [7,8].

After determining the best sample $\mathcal{X}^\ell + \Delta\mathcal{X}^\ell$ with respect to the vector $\mathcal{X}^{\ell+1} = \mathcal{X}^\ell + \Delta\mathcal{X}^\ell$, a series of m trials $\mathcal{X}^{\ell+1} + \xi$ where ξ is a random vector, $(\xi_1, \xi_2, \dots, \xi_\kappa)$ normally distributed with respect to $\Delta\mathcal{X}^\ell$ with variance $\delta = (\delta_1, \delta_2, \dots, \delta_\kappa)$.

The variance depends on the rate of change of the non-variance function

$$(4) \qquad \delta^{\ell+1} \begin{cases} 0{,}5\, \delta^\ell, if\ \Delta Q^\ell < \Delta Q^{\ell-1} \\ 2\delta^\ell,\ same\ \Delta Q^\ell \geq \lambda \Delta Q^{\ell-1} \end{cases}$$

where ΔQ^ℓ and $\Delta Q^{\ell-1}$ are the inconsistency increments for the ℓ-th and $(\ell-1)$ steps, respectively: $\mathcal{P}(\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_\kappa)$ is the vector bounding δ from above; $\mathcal{H}(\hbar_1, \hbar_2, \dots, \hbar_\kappa)$ is the initial value of δ.

From the series of trials, the best $\mathcal{X}^{\ell+1} + \Delta\mathcal{X}^{\ell+1}$ is selected, and a series of m normally distributed trials is performed again, and so on.

When $\Delta Q^\ell > 0$, we proceed to a series of m uniformly distributed trials.

When several series of m uniformly distributed trials are unsuccessful, i.e. do not reduce $Q(x)$, the step is halved and new series of trials are performed until the condition of the local minimum is fulfilled.

$$\mathcal{K}_{\mathcal{H}} = \mathcal{K}_0{}^{\downarrow} \mathcal{E} < \Delta Q^*, \quad (5)$$

where $\mathcal{K}_{\mathcal{H}}$-number of consecutive unsuccessful series of trials; $\mathcal{K}_0 = const$ determines the browsing density of the neighborhood of the local minimum: $\mathcal{E}$-finding accuracy; ΔQ^*-the largest increment over the last unsuccessful series.

To create a control system for multistage flotation processes it is advisable to study the characteristics of individual circuits. The latter include main, control, pre-flotation, pro-product and recleaning. For each circuit in (6) were determined optimal values of their output parameters, providing maximum concentrate yield for the process.

$$\theta_{moul} = \beta_{10} + \frac{\alpha - \beta_{10}}{1 - \gamma_{10}}, \qquad (6)$$

The task of circuit optimization is to maintain the concentrate output of each circuit in the specified values by varying the control parameters of the process in the permissible area and meeting certain constraints taken from Table 2.

When solving this problem, we use the mathematical models given in Table 2 [11].

Construction of a micromodel of separate circuits of MSP flotation is necessary to determine the influence of input perturbing and control parameters on the output indicators of each circuit.

In order to build a mathematical model of the individual circuits of MSP flotation, an active experiment was conducted in real production conditions at the eighth section of the copper and ore dressing plant of NMMK.

When planning the experiment, the main level was chosen taking into account the existing technological regime at the factory. The total reagent costs were emphasized at the levels set by the planning matrix and automatic control systems. Before the start of the experiment, the pulp level in the sump and in all flotation machines of the section under study stabilized. During the active experiment, the control and disturbing parameters were measured. Reagents were supplied at intervals to the input of each circuit, and samples were taken manually at the output of each circuit. Forty-hour observations were carried out on a normally functioning section of the multistage flotation process (Table 1.).

Table 1
Planning matrix and experimental results

Experience Number	Contours and factors										Optimization parameters, %				
	I			II		III		IV		V	β_2	β_4	β_6	β_8	β_{10}
	X_1	X_2	X_3	X_1	X_2	X_1	X_2	X_1	X_2	X_1					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	-	-	-	+	-	-	-	+	+	+	5,97	5,16	5,76	4,00	20,15
2	-	-	-	+	-	-	-	+	+	+	11,83	5,12	5,76	3,36	21,83
3	-	-	-	+	-	-	-	+	+	+	10,94	5,44	4,96	3,20	20,94
4	-	-	-	+	-	-	-	+	+	+	6,57	5,50	2,68	3,20	19,80
5	-	-	-	+	-	-	-	+	+	+	4,07	4,80	5,60	4,16	20,17
6	+	-	-	-	+	+	+	+	-	-	8,59	2,80	2,80	7,86	18,59
7	+	-	-	-	+	+	+	+	-	-	12,57	3,84	6,80	7,12	18,11
8	+	-	-	-	+	+	+	+	-	-	8,67	3,44	3,12	4,96	19,73
9	+	-	-	-	+	+	+	+	-	-	9,06	3,84	7,14	4,16	20,15
10	+	-	-	-	+	+	+	+	-	-	8,99	5,28	3,76	4,08	19,15
11	+	+	-	+	+	-	+	-	-	-	14,62	3,80	6,76	3,04	22,46
12	+	+	-	+	+	-	+	-	-	-	8,71	1,44	7,96	4,56	20,18
13	+	+	-	+	+	-	+	-	-	-	7,40	3,96	6,28	3,36	21,70
14	+	+	-	+	+	-	+	-	-	-	15,76	5,20	4,52	2,24	19,47
15	+	+	-	+	+	-	+	-	-	-	11,86	3,04	6,28	4,16	18,52
16	-	+	-	-	-	+	-	-	+	-	10,98	2,01	3,32	1,28	19,09
17	-	+	-	-	-	+	-	-	+	-	9,38	1,68	3,46	1,28	17,14
18	-	+	-	-	-	+	-	-	+	-	9,95	2,24	2,96	1,12	19,03
19	-	+	-	-	-	+	-	-	+	-	10,40	2,83	6,62	1,28	19,15
20	-	+	-	-	-	+	-	-	+	-	12,11	2,72	5,32	1,12	18,90
21	-	+	+	-	-	-	-	+	-	+	9,12	2,96	4,32	6,88	22,53

The intervals between observations were three minutes.

When building a micromodel of the flotation process, the following most significant parameters were identified for each of the circuits:

I main:

y- the content of valuable metal (copper) in the concentrate of the main flotation $\beta_2, \%, x_1$- the consumption of sodium sulfide Na_2S, g / t; x_2- the same xanthate KSt, g/t; x_3- the same foaming agent T-80; g/t; x_4- content of valuable metal in food α, %; x_5- pulp alkalinity pH, %; x_6- productivity, Q, %; x_7- pulp density in the feed π_{π_α}, %; x_8- the same in the dump discharge of the main flotation $\pi_{\pi_{\beta_1}}$, %; x_9- the content of valuable metal in the final product of the

Table 1
Models of flotation process contours in relation to copper content in concentra

Contour	Model
In real scale (a)	
Basic	$y = \beta_2$=76,7813+0,0057x_1-0,0399x_2-0,1863x_3-37,1979x_4+0,8053x_5 -0,0076x_6 +12,4656x_7 -44,0540x_8 +19,9862x_9 -0,0100x_{10}
Control	$y = \beta_4$=0,5876+0,3066 x_1, -0,0561x_2 -27,8452x_3 +3,0077x_4 -1,3284x_5+8,4164x_6 -0,I863x_7 -0,0382x_1x_2 +5,5642x_2x_3
Before flotation	$y = \beta_6$=10,5076+0,0831x_1+0,4483x_2+57,5891x_3+4,2084x_4-11,9817x_5+5,5816x_6-0,3798x_7-0,0163x_1x_2 -3,4223x_2x_3
Industrial	$y = \beta_8$=79.0377+0,1018x_1-1,3535x_2-2,2275x_3 -3,8305x_4 +3.5253x_5 -I,I070x_1 +77,5I82x_1 +0,5296 x_1x_2+0,3371x_2x_3
Industrial	$y = \beta_{10}$=5965+0,0032x_1 -0,0562x_2 -0,0560x_3 -0,1662x_4 -1,9177x_5-26,7642x_1x_2 +0,1272x_2x_3
On a standardized scale (b)	
Basic	$y = \beta_2$= 0,0391x_1-0,1104x_2-0,3218x_3 -0,5533$x_4$0,1373x_5 -0,0457x_6 +0,1335x_7 -0,5576x_8+0,2315x_9 -0,3309x_{10}
Control	$y = \beta_4$= 1,1179x_1, -0,6184x_2--0,6725x_3+0,0803x_4-0,0543x_5+0,1484x_6-0,0670x_6 -1,0414x_1x_2 +1,2752x_2x_3
Before flotation	$y = \beta_6$= 0,1299x_1 +1,3899x_2- +1,0768x_3 +0,I286x_4 -0,5202x_5 +0,0588x_6-0,I452x_7 -0,4537x_1x_2 -1,5628x_2x_3
Industrial	$y = \beta_8$= 0,3075x_1-I,3625x_2-0,3759x_3-0,I5584x_4+0,0409x_5-0,0122x_6+0,5309x_7 +0,1353x_1x_2 +1,9609x_2x_3
Industrial	$y = \beta_{10}$=0,3797x_1 -0,1112x_2-0,0494x_3-0,2234x_4-0,1010x_5 -0,4016x_6 +0,0428x_7

main flotation β,%; x_{10}- volume flow of pulp in the main flotation feed, m3/h;

II control:

y- the content of valuable metal in the concentrate of the control flotation β_4, %; x_1 —sodium sulfide consumption $Na_2\delta$ g/t; x_2- the same xanthate KSt,g/t; x_3- the content of valuable motall in the final product of the main flotation β_1, %, x_4- the density of the panels in the dump discharge of the main flotation $\pi_{\pi_{\beta_1}}$, , % x_5- the same control flotation $\pi_{\pi_{\beta_3}}$, x_6- the content of valuable metal in the final product of the control flotation β, %; x_7- pulp alkalinity;

III pre-flotation:

y- content of valuable metal in the pre-flotation circuit concentrate β_6,%; x_1- xanthate consumption KSt, g/t; x_2- the foaming agent T-80, g/t; x_3- content of valuable metal in the waste product of the control circuit β_3, %; x_4- the same pre-photo contour β_5, %; x_5- pulp density in the dump discharge of the control circuit $\pi_\pi\beta_3$, %; 7; x_6- the same pre-flotation circuit $\pi_\pi\beta_5$, %; x_7- pulp alkalinity,pH

I V middling:

y- the content of valuable metal in the concentrate of the middling circuit β_8,%; x_1- xanthate consumption KSt, g/t; x_2– the same foaming agent T-80, g/t; x_3- the content of ferrous metal in the control flotation concentrate β_6, %; x_4- the same in the final product of the cleaning circuit β_8, %; x_5- the same in the dump product of the middling circuit β_9, %; x_6- pulp density in the dump discharge of the enumerated contour $\pi_\pi\beta_9$, %;x_7 the same middling contour $\pi_\pi\beta_7$, %; x_8- pulp alkalinity, pH.

V cleaner:

y- the content of valuable metal in the concentrate cleaning circuit β_{10}, %; x_1- lime consumption CaO, g/t; x_2- the content of valuable metal in the concentrate of the main circuit β_2,%; x_3- the same deflotation circuit β_6, % x_4- the same middling circuit β_8, %; x_5- the same in the final product of the cleaning circuit β_9, %; x_6- pulp density in the final discharge of the cleaning circuit $\pi_\pi\beta_4$, %; x_7- pulp alkalinity, pH.

The pulp temperature is in the range of 8 ÷10 ^{0} C. In this case, in all cases, the sub-metal content means the copper content. When constructing a model of the MSP flotation circuit, the copper content in the ore, the consumption of reagents (xanthate, foaming agent, sodium sulfide, lime), alkalinity of the pulp and its density, productivity and temperature are taken as input parameters; it and in waste products.

The study of the dependences of the main indicators of the process showed that the copper content in the concentrate and in the waste, products depend on the consumption of the reagent. These dependences are admissibly approximated by a polynomial no higher than the second degree, and the extremum in the region of optimal reagent consumption values suggests the possibility of extreme regulation.

Based on the results of planning the experiment, the regression analysis method was used to construct mathematical models for individual circuits of the flotation process, both in natural and in a standardized scale (Table 2). To obtain a statistical model of the process under study, the library of standard PC programs was used; with its help, a program was compiled for calculating the coefficients of the model and analyzing the parameters included in the model.

Mean values and dispersion of parameters characterizing. dispersion of experimental results is determined by formulas (7) and (8).

$$R_{Kp} = m_E + \sum_{i=1}^{n} \xi_i K_E(\tau_i) \qquad (7),$$

$$R_{Kp} = m \int_0^T W(\tau) d\tau - m_y + \sum_{i=1}^{n} \xi_i \left[\int_0^T \int_0^T K_x \ldots . (\tau_i - \lambda + \tau) W(\tau) d\tau d\lambda - 2 \int_0^T K_{xy} (\tau_i + \tau) W(\tau) d\tau + K_y(\tau_i) \right], \quad (8)$$

The adequacy of the obtained equations was checked using the Fisher criterion [3, 11]:

$$F = \frac{S_{ag}^2}{S_y^2} \le F(0{,}005;\ f_{ag}; f_y),$$

Where S_{ag} - variance of the adequacy of the model, determined by the formula

$$S_{ag} = \frac{\sum_{i=1}^{n}(y_i - y_i^*)^2}{n - m - 1}$$

here y_i- the calculated value of the quantity in the i -th experiment; $\mathcal{F}(0{,}005;\ f_{ag}; f_y)$- Fisher's test at 5% significance level; $f_{ag}= m - I$ –- number of degrees of freedom of the dispersion of adequacy; f_y- the same reproducibility; n- the same experiments: $m$the same parameters.

The minimum control costs are chosen as the optimization criterion

$$S = \sum_{\kappa=1}^{n} C_\kappa U_{i\kappa} \to min \quad (9)$$

at maintaining the content of useful metal in the concentrate at the outlet of each at the specified values

$$\beta_{2i\ give} - \beta_{2i}(x_{ij}, u_{i\kappa}, \theta_{i\ell}) = 0 \quad (10)$$

and constraints in the form of inequalities

$$\theta_{i\ell} \geq \varphi_i(x_{ij}, u_{i\kappa}, \gamma_{is}, \beta_{2i}), \qquad (11)$$

as well as bilateral constraints on the variables

$$x_{ij}^- \leq x_{ij} \leq x_{ij}^+, \qquad u_{i\kappa}^- \leq u_{i\kappa} \leq u_{i\kappa}^+,$$
$$\gamma_{is}^- \leq \gamma_{is} \leq \gamma_{is}^+, \qquad \gamma_{is} > 0, x_{ij} \geq 0, u_{i\kappa} \geq 0, i = \overline{1,n}, \qquad (12)$$

where C_κ -cost of κ -th control parameter, i - loop number, κ, j, s - numbers of control, input parameters and solid consumption, respectively, $\beta_{2i\ giv}$-determined quality of the output indicator, $u_{i\kappa}$- control parameters (consumption of sodium sulfide, butyl xanthogenate, foaming agent, lime), x_{ij}-input parameters content of useful metal in feedstock, pulp alkalinity, productivity, volume flow rate, γ_{is}- consumption of solid, β_{2i}, φ_i-structure of the mathematical model and constraints of the i - control loop.

The optimization problem for the whole technological scheme has the form:

$$C = \sum_{i=1}^{r}\sum_{\kappa=1}^{n} C_\kappa U_{i\kappa} \to min \quad (13)$$

or

$$C = \sum_{i=1}^{r} C_i \quad (\ r\text{- number of circuits}) \ (14)$$

if the conditions are met (11), (12).

Application of the random search algorithm to select the optimal values of control parameters of the process varying in the given area $\mathcal{D}$ allows us to obtain the coordinates of reagents (control parameters) that give local and global minima for the control cost.

Extreme values of the functions $\beta_{i\ calc.}$ and the corresponding reagent costs (table 2), which ensure fulfillment of conditions (11), (12), were determined for the efficiency criteria by the random search method on the CT, as shown in Table 1.

RESULTS ANS DISCUSSION. Thus, with the help of the optimization algorithm we strive to the set value found when solving the inter-loop optimization problem. Maintaining the quality indicator of the output concentrate within the specified values is carried out by selecting a vector of controllable parameters (reagents) that satisfy the imposed constraints.

Application of the random search method allowed to determine the optimal reagent mode, which is necessary to create a system of optimal control of MSP of copper ore flotation.

To evaluate the effectiveness of the obtained results, a comparative analysis was carried out with the current values of control parameters in the conditions of a normally functioning object for a one-month period. The obtained data after appropriate treatments are summarized in table 3.

In addition to the main control parameters during the period of control check such reagents as aeroflot and spindle oil were fed in the process. The controlled parameters were maintained within the limits: $p\mathcal{H}$-alkalinity of pulp - 9÷9,5, density mode of flotation: main flotation feed - 28÷30% of solid, promproduct flotation - 15+18% of solid, second recleaning feed - 18÷20% of solid, pulp temperature -- 8÷12^0 C.

Fig. 1 shows the distribution of flotation process control costs by contours. The graph shows that the control costs using contour optimization (dashed line) compared to the current costs at the plant without optimization (continuous line) in the main, control, intermediate and clean flotation are lower, and significantly higher in pre-flotation.

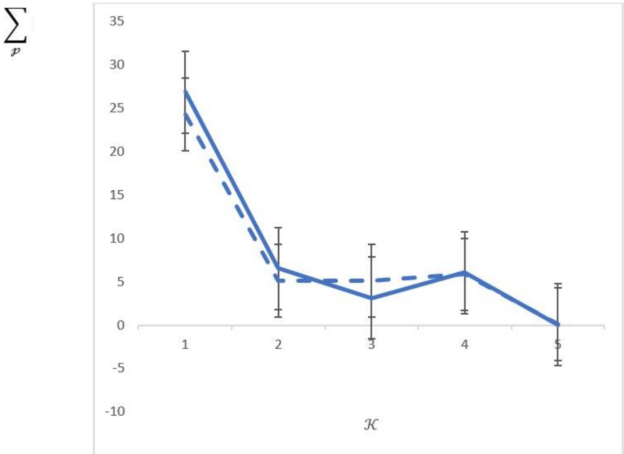

Fig. 1. Comparative control costs by loop, where $\sum_{\wp}$ is the total reagent flow rate, $\mathcal{K}$ is the loop numbers.

Table 2
Values of control parameters after solving contour optimization problems

Factors	Units of measurement	Control contours					Total consumption
		BF	CF	BF	IF	IF	
Sodium sulphide	$\frac{kg/10^3t}{sum}$	$\frac{33,341}{2,54}$	$\frac{10,507}{7,92}$	-	-	-	$\frac{43,848}{10,460}$
Butyl xanthogenate	$\frac{kg/10^3t}{sum}$	$\frac{13,759}{12,37}$	$\frac{2,880}{2,58}$	$\frac{3,335}{2,99}$	$\frac{5,452}{4,90}$	-	$\frac{25,426}{22,840}$
Foamer	$\frac{kg/10^3t}{sum}$	$\frac{18,567}{3,97}$		$\frac{9,626}{2,08}$	$\frac{4,873}{0,95}$	-	$\frac{33,066}{7,000}$
Lime	$\frac{kg/10^3t}{sum}$	-	-	-	-	$\frac{0,773}{0,008}$	$\frac{0,773}{0,008}$
$\sum_{\wp}$	sum	24,26	5,12	5,07	5,85	0,008	40,308
β_{2i}	%	9,893	3,430	4,254	5,270	20,000	

Nevertheless, the redistribution of costs between circuits is economically justified for the concentrator as a whole: the total management costs are reduced per thousand tons of ore processed, which is 5.24%. At the same time, an increase in concentrate quality is observed.

The outlined algorithm of contour optimization of the flotation process has a number of advantages: a small amount of machine memory, fast performance and relatively easy implementation in production conditions.[11]

Table 3
Values of control parameters without optimization

Factors	Units of measurement	Control contours					Total consumption
		BF	CF	BF	IF	IF	
Sodium sulphide	$\frac{kg/10^3t}{sum}$	$\frac{40,0}{9,509}$	$\frac{18,0}{4,279}$	-	-	-	$\frac{58,0}{13,788}$
Butyl xanthogenate	$\frac{kg/10^3t}{sum}$	$\frac{14,0}{12,586}$	$\frac{2,5}{2,248}$	$\frac{3,0}{2,697}$	$\frac{6,0}{5,394}$	-	$\frac{25,5}{22,925}$
Foamer	$\frac{kg/10^3t}{sum}$	$\frac{22,0}{4,739}$		$\frac{2,0}{0,431}$	$\frac{3,0}{0,647}$	-	$\frac{27,0}{5,817}$
Lime	$\frac{kg/10^3t}{sum}$	-	-	-	-	$\frac{1,546}{0,016}$	$\frac{1,546}{0,016}$
$\sum_{\wp}$	sum	26,834	5,527	3,128	6,041	0,016	42,546
β_{2i}	%	8,73	3,36	3,91	5,18	18,73	

CONCLUSION. In conclusion, the local contour optimization algorithm is a powerful tool for improving the efficiency of multistage flotation processes. This approach allows systematic optimization of process parameters in real time, taking into account the variability of input data and the quality requirements of the final product. Based on the local contour optimization algorithm, automated control systems can be developed that can adapt to changes in production conditions and achieve optimal results.

However, it should be taken into account that successful implementation of the algorithm requires accurate calibration of process models and careful control of optimization parameters. In addition, the integration of the algorithm into real production systems may require significant efforts to train personnel and modernize the technical infrastructure.

Overall, the local contour optimization algorithm represents a promising area of research in flotation, which can lead to significant improvements in the efficiency and economic viability of ore beneficiation processes. Further research and development in this area may lead to new methods and tools to optimize flotation processes and improve the competitiveness of companies in the market.

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